

Seeing Is Deceiving

● Yellow Level: Ready for a Challenge

Name: _____ Date: _____ Period: _____

The Science of Seeing

Your eyes collect light, but your brain decides what you're seeing. Your brain uses shortcuts to process information quickly—but these shortcuts can be fooled. That's why optical illusions work!

Key Concept: Perception isn't passive—your brain actively constructs reality based on patterns, expectations, and context. Artists have used this for centuries to create depth, movement, and impossible images.

Activity 1: Analyze an Illusion

Examine the optical illusion shown in class. Sketch it below and analyze how it works:

Sketch the illusion

Analysis Questions:

What visual trick is being used? (contrast, perspective, pattern, context, etc.)

What assumption is your brain making that causes the illusion?

How might an artist use this technique intentionally?

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Activity 2: Design Your Own Optical Illusion

Create an original optical illusion using one of these techniques:

- **Ambiguous figures:** Images that can be seen as two different things
- **Size/perspective illusions:** Objects that appear different sizes due to context
- **Motion illusions:** Static patterns that appear to move
- **Impossible objects:** Shapes that couldn't exist in 3D space

Design your optical illusion here

Explain your design:

What technique did you use?

What assumption does your illusion exploit?

Mini Museum Connection

The Victorian era (1837-1901) saw an explosion of optical toys and illusions. Visit minimuseumproject.com/exhibitions/seeing-is-deceiving to see how 19th-century inventors used these same principles!